

Section A

Answer **all** the questions in this section.

- 1 Fig. 1.1 below shows 2 organelles from an animal cell.

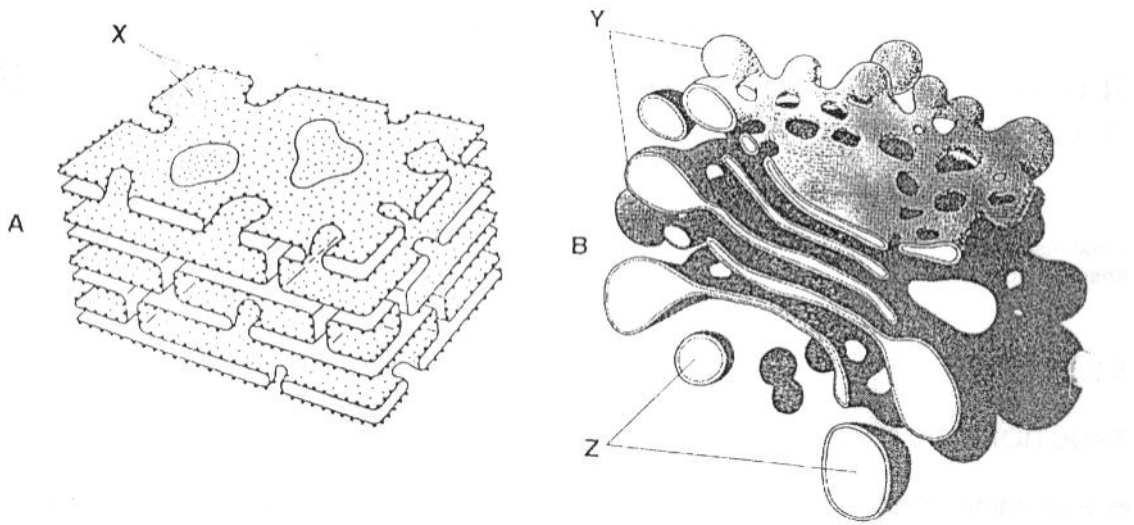


Fig. 1.1

- (a) (i) Name the Organelles A and B

A

B

[1]

- (ii) Underline the cells in the following list that do not have any A.

liver cell; bacterium; virus; muscle fibre

[1]

- (iii) Name the structures Y and Z shown in Fig. 1.1B.

Y

Z

[1]

- (b) Explain the function of A and B with relation to protein synthesis.

[4]

[Total: 7]

2 Fig. 2.1 illustrates the binding of a protein with a DNA molecule.

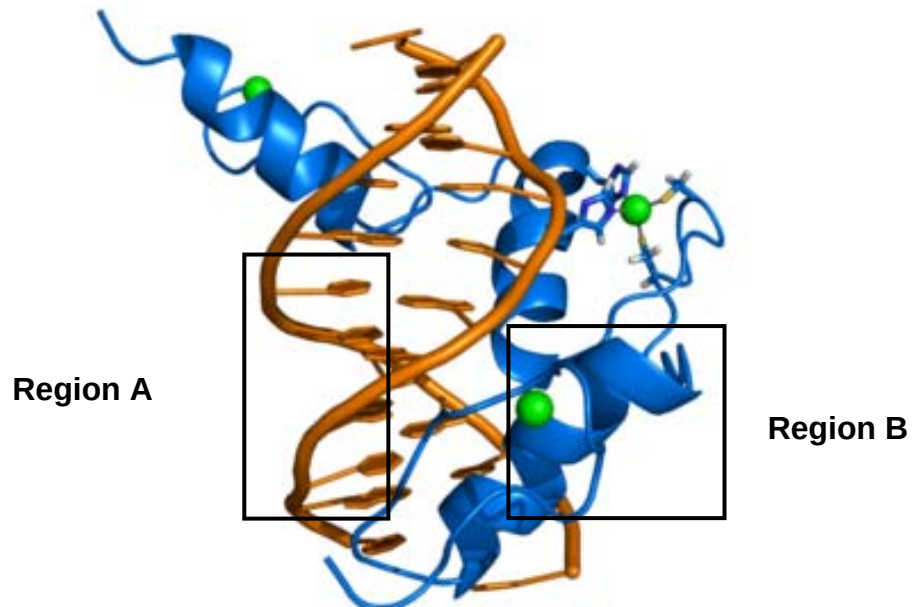


Fig. 2.1

(a) With reference to Fig. 2.1,

(i) Describe in what ways the helical structure show in Region A and B differ.

[3]

Region B contains a zinc finger protein motif. It is a highly conserved protein motif and the binding specificity is conferred by a short stretch within the α -helix domain. Transcription factors often contain zinc finger motifs.

(ii) Suggest why a transcription factor only binds to certain part of a DNA molecule.

[3]

(b) (i) State what is a control element.

[1]

- (ii) Explain how a control element can increase gene expression.

[2]

[Total: 9]

- 3 Fig. 3.1 below shows the movement of O_2 and CO_2 between two organelles, A and B, in a mesophyll cell of a leaf, in the presence of light.

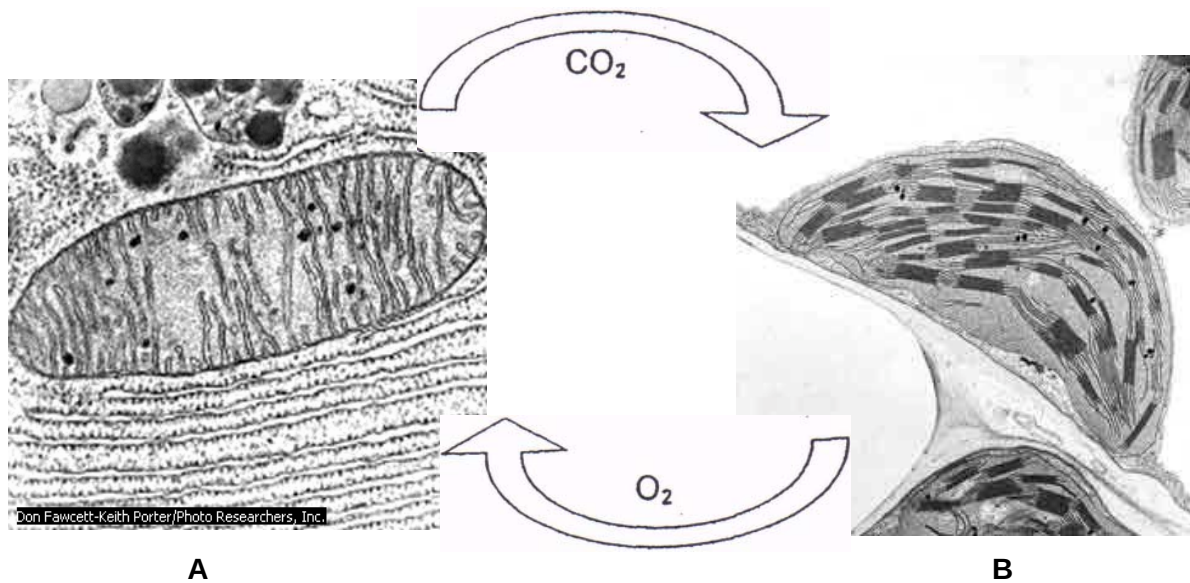


Fig. 3.1 Movement of O_2 and CO_2 between organelles A and B

- (a) State the process taking place in:

organelle **A**:

organelle **B**:

[1]

- (b) Briefly describe how organelle B is structurally adapted to carry out its function/s.

[3]

- (c) ATP synthesis takes place in both organelles A and B. State three ways in which the process of ATP synthesis differs in the two organelles.

[3]

[Total: 7]

- 4 A variety of oilseed rape shows variation in two characters: stem height and chlorophyll distribution in its leaves. Plants may be tall or short. Chlorophyll distribution may be uniform, producing normal green leaves; patchy, producing variegated leaves; or totally absent. Plants in which chlorophyll is totally absent die before they emerge above ground as seedlings.

Pure breeding plants with tall stems and normal green leaves were crossed with plants with short stems and variegated leaves. The generation from this cross consisted of plants with tall stems and green leaves and plants with tall stems and variegated leaves in a ratio of 1:1.

- (a) Using suitable symbols, construct a diagram of this cross to show the genotypes of the parents and the offspring produced.

[3]

- (b) The first generation plants with tall stems and variegated leaves were self-pollinated and 1000 seeds sown. Six weeks after germination, the phenotypes of the second generation plants were recorded. The following results were obtained:

Plants with tall stems and green leaves	190
Plants with tall stems and variegated leaves	386
Plants with short stems and green leaves	61
Plants with short stems and variegated leaves	126

Construct a diagram to show the results of the self pollination.

- 5 Restriction enzymes are used during genetic engineering. The target sites for *EcoRI*, *BamHI* and *AluI* is shown in Fig. 5.1 below.

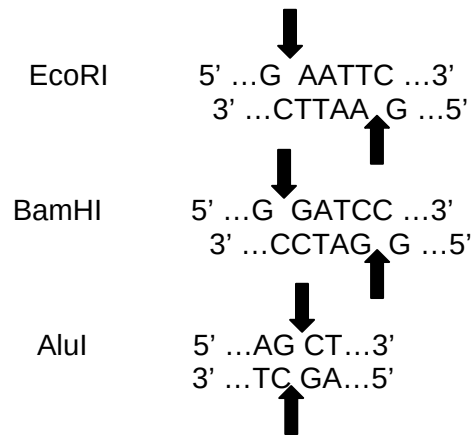


Fig. 5.1

A bacterial plasmid is shown in Fig.5.2 below. There are no target sequences for the three restriction enzymes in the rest of the plasmid.

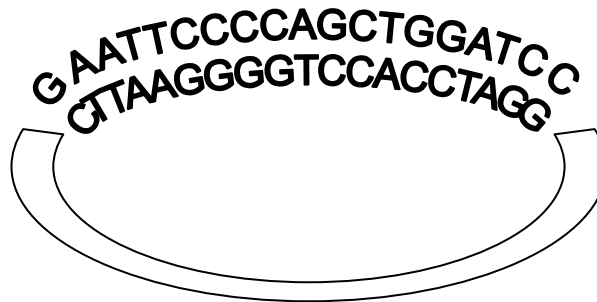


Fig. 5.2

With reference to **Fig. 5.1** given above,

- (a) State how many fragments of DNA will be present after the plasmid has been cut with
- (i) a mixture of *EcoRI* and *BamHI*

- (ii) a mixture of *EcoRI* and *AluI*

[2]

- (b) A student incubates the plasmid molecule shown in Fig. 5.2 with *EcoRI* to insert a human gene. Outline how restriction enzymes may be used in the formation of recombinant DNA molecules.

[3]

- (c) Name two other enzymes used in the formation of recombinant DNA and briefly describe their roles.

[2]

- (d) Gel electrophoresis is usually performed after restriction digestion to separate DNA fragment. Briefly describe how DNA fragments may be separated by gel electrophoresis.

[3]

[Total: 10]

Section B

Answer **EITHER** 6 or 7

- 6 (a) Describe the structure and function of portions of eukaryotic DNA that do not encode for protein or RNA. [10]

- (b) Describe how a gene may be transferred into a named bacterium to obtain large amounts of useful proteins. [10]

[Total : 20]

- 7 (a) Discuss the benefits and ethical concerns of the Human Genome Project. [10]

- (b) Describe the different types of stem cells and their functions. [10]

[Total : 20]

END OF PAPER